EVision – Electric Vehicle Charging Demand Analytics

Final Project Review | Department of Computer Science Engineering | KL University

TEAM MEMBERS

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Academic Year: [2026–27]

Introduction & Problem Statement

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Background

Electric vehicle adoption is increasing, creating a growing need for efficient charging infrastructure. Understanding charging patterns is important for managing demand and improving charging-station utilization.

Problem Statement

Unpredictable EV charging demand can cause peak-hour congestion and under-utilization of charging stations. **EVision** aims to analyze historical charging data and predict future demand to support efficient capacity planning.

Motivation / Need

- Helps charging-station operators understand demand patterns.
- Supports better utilization of charging infrastructure.
- Reduces uncertainty in charging-capacity planning.



Objectives & Scope

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Objectives

01	**Collect and Preprocess Data**	Collect historical EV charging-session data and clean missing, duplicate, and inconsistent records to prepare a reliable dataset for analysis.
02	**Analyze Charging Patterns**	Examine charging demand across different time periods and stations to identify peak-demand periods and variations in charging behaviour.
03	**Develop a Prediction Model**	Develop a machine-learning model to predict future EV charging demand using relevant historical charging features.
04	**Evaluate and Visualize Results**	Evaluate prediction performance using MAE, RMSE, and R^2, and present charging insights through an interactive dashboard.

Project Scope

- Historical EV charging-data analysis.
- Data preprocessing and exploratory analysis.
- Machine-learning-based demand prediction.
- Interactive dashboard and visualization.

Literature Survey

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S.No.	Author(s) & Year	Method / Technique	Key Findings	Limitation / Gap
1	Orzechowski et al. (2023)	Supervised and deep-learning models for medium-term public charging demand forecasting	Forecasts demand up to one week ahead across multiple stations; reported network-demand SMAPE of 5.9%.	Evaluated using 11 public stations in Scotland; performance depends on dataset coverage and forecast horizon.
2	Wang et al. (2023)	LSTM for station-level short-term forecasting (1–5 hours)	LSTM outperformed ARIMA and MLP on a Beijing EV trajectory dataset.	Study is based on a specific city and short-term forecasting setup.
3	Alam et al. (2024)	Multivariate forecasting; compares GRU, LSTM, Random Forest and other models	Multivariate inputs improved forecasting; GRU achieved the best reported multivariate results.	Evaluation uses a specific historical dataset and aggregation approach; broader regional validation is needed.
4	Wang et al. (2026)	Comparison of six ML/DL methods across 12 real-world scenarios	Traditional ML remained competitive and models showed geographical transferability.	Performance can vary with regional charging patterns and data availability.

Research Gap

Existing studies demonstrate effective EV charging-demand forecasting, but results are often tied to particular datasets, locations, and forecasting horizons. EVision brings historical charging-data analysis, station-level demand insights, prediction, and an interactive dashboard together for capacity-planning support; its Hyderabad applicability must be validated with representative local charging data.

Proposed System & Methodology

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1 Define Problem

- Analyze the need for EV charging demand forecasting
- Define scope and objectives
- Identify success criteria

2 Collect Data

- Gather data from CSV/Excel, databases and APIs
- Collect historical charging-session data
- Include station details, location and time information

3 Design

- Design system architecture
- Select algorithms (Random Forest)
- Choose tools and technologies
- Define data flow and module structure

4 Implement

- Develop data preprocessing pipeline
- Build prediction model
- Implement backend (FastAPI) and frontend (React + TypeScript)
- Integrate database (MongoDB)

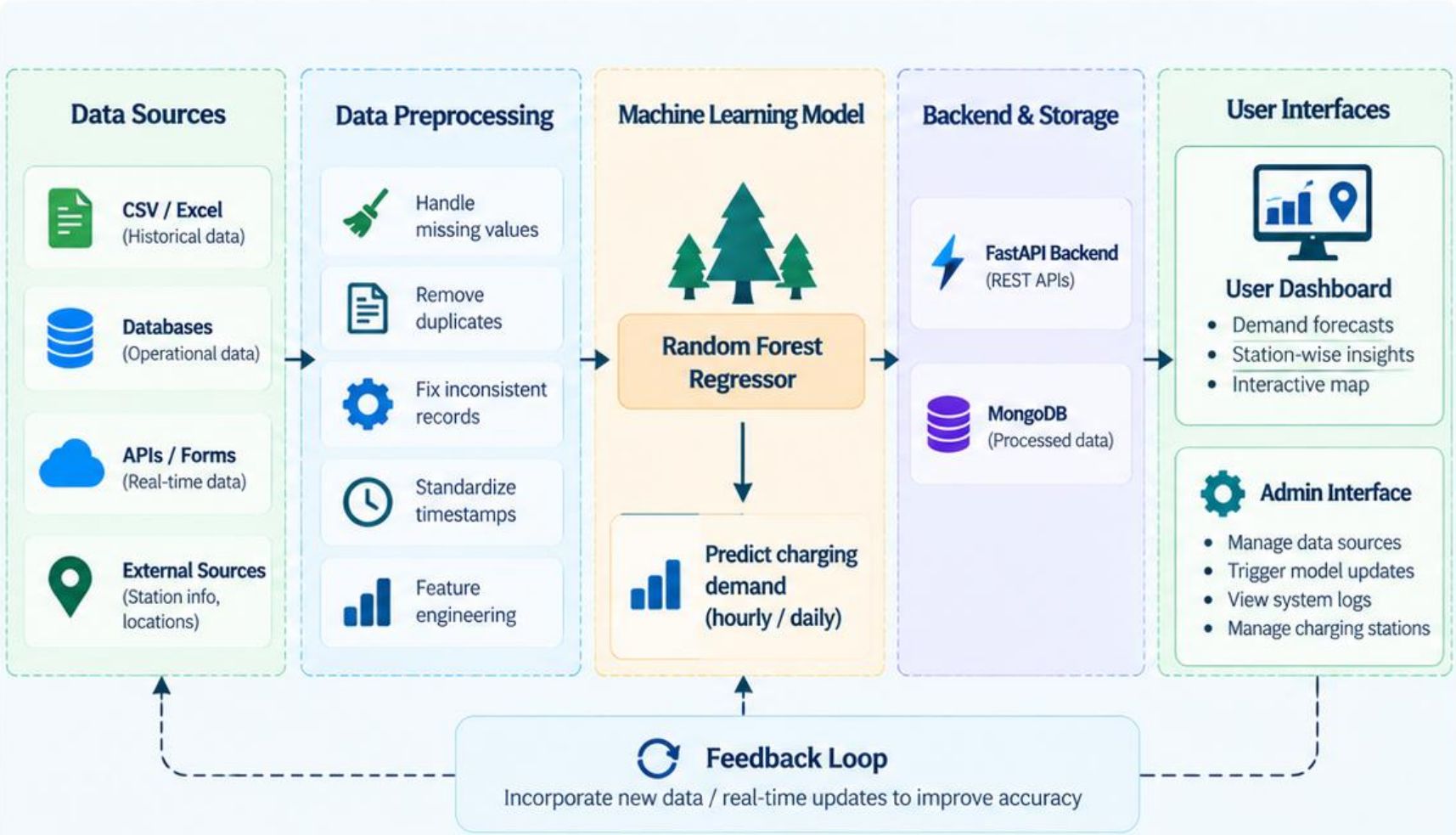
5 Test & Evaluate

- Evaluate model using MAE, RMSE and R^2 score
- Test with real-world scenarios
- Validate system functionality
- Compare results and analyze performance

Key Novelty / Contribution

- Integrates multi-source EV charging data with engineered temporal and spatial features to improve demand prediction accuracy.
- Provides an interactive dashboard with station-wise insights and map visualization for better infrastructure planning.

System Architecture



Components

1 Data Sources
Collect charging data from CSV/Excel files, databases and APIs (historical, operational and real-time).

2 Data Preprocessing
Clean data, handle missing values, remove duplicates, fix inconsistencies, standardize timestamps and create relevant features.

3 Prediction Model
Use Random Forest Regressor to predict charging demand (hourly / daily) based on historical and engineered features.

4 Backend, Database & Dashboard
FastAPI provides REST APIs, MongoDB stores processed data, and a React web dashboard visualizes forecasts, trends and station-wise insights for users and admin.

Hardware & Software Requirements

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Hardware

- Intel Core i5 / equivalent processor
- 8 GB RAM or higher
- 256 GB SSD or higher
- Stable internet connection

Software & Tools

- Windows 10 / 11
- Google Antigravity IDE
- Git and GitHub

Libraries / Frameworks

- FastAPI (REST API)
- React (Frontend)
- MongoDB
- Scikit-learn (Random Forest)

Deployment / Platform: Google Antigravity IDE | Web Browser (Application Launch)

Dataset / Data Understanding

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18,642
Charging Sessions (Hyderabad only)

14
Features (columns)

0
Missing values

2
Charger types (AC / DC)

- **Source**: Hyderabad EV charging station data (Open data / publicly available dataset)
- **Key features**: station ID, location (Hyderabad areas), charger type, start time, duration, energy demand, and other session details.
- **Data quality**: No missing values, timestamp formats consistent, and limited duplicates (removed during preprocessing).

Sample records (Different locations in Hyderabad)

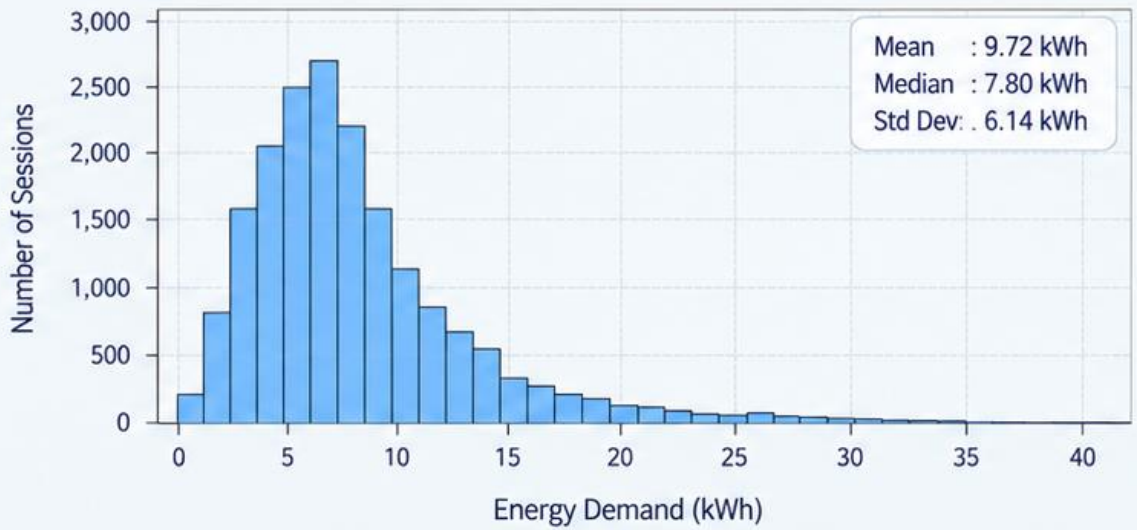
Session ID	Location (Hyderabad)	Charger Type	Start Time	Duration (hours)	Energy Demand (kWh)
HYD_0001	HITEC City	AC	2023-07-01 08:15:00	1.20	10.40
HYD_0002	Gachibowli	DC	2023-07-01 10:32:00	0.75	18.60
HYD_0003	Madhapur	AC	2023-07-01 12:10:00	1.05	9.20
HYD_0004	Kondapur	DC	2023-07-01 14:45:00	1.80	24.50
HYD_0005	Jubilee Hills	AC	2023-07-01 16:20:00	0.90	8.30
HYD_0006	Banjara Hills	AC	2023-07-01 18:10:00	1.40	12.10
HYD_0007	Secunderabad	DC	2023-07-01 19:25:00	0.65	15.70
HYD_0008	Uppal	AC	2023-07-01 21:05:00	1.10	11.90
HYD_0009	LB Nagar	DC	2023-07-01 22:30:00	0.80	17.40
HYD_0010	Kukatpally	AC	2023-07-02 07:50:00	1.25	10.80

Data Preprocessing & Exploratory Analysis

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Energy Demand Distribution

Distribution of energy demand per charging session in Hyderabad.



Energy Demand by Location (Hyderabad)

Comparison of energy demand across different locations in Hyderabad.



Cleaning

- Checked and handled missing values (No missing values found).
- Removed duplicate sessions (0.8% removed).
- Validated charger IDs, locations and timestamps.
- Handled outliers in energy demand using IQR method.

Transformation

- Converted timestamps to hour, day, month and day of week.
- Encoded categorical features (location, charger type).
- Scaled numerical features using StandardScaler.
- Created time-based and location-based features for better analysis.

Feature Selection

- Selected key features based on domain knowledge and analysis.
- Final features: location, charger type, start hour, duration, day of week, and energy demand (kWh).
- Removed low-variance and redundant features.

Module-wise Description

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1 Data Collection & Preproessing

Purpose: Collect and prepare reliable EV charging session data for analysis and modeling.

Input → Output: Raw charging station records (location, time, charger type, session details) → Cleaned and structured dataset (removed missing values, duplicates, handled outliers).

Developed by: Project team

2 Exploratory Data Analysis

Purpose: Understand charging demand patterns across different Hyderabad locations and time periods.

Input → Output: Cleaned dataset → Visual insights (hourly demand, daily/weekly trends, location-wise comparison, charger type distribution).

Developed by: Project team

3 Demand Prediction Model

Purpose: Predict future EV charging demand using a machine learning model (Random Forest Regressor).

Input → Output: Processed features (time, location, charger type, etc.) → Predicted energy demand (kWh) for future time periods and locations.

Developed by: Project team

4 Dashboard & Decision Support

Purpose: Provide an interactive web application to visualize demand insights and support charging infrastructure planning.

Input → Output: Predicted demand and analysis results → Interactive dashboards (charts, maps, location insights, planning recommendations).

Developed by: Project team

Algorithms & Technologies Used

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Random Forest Regressor

Why chosen

Handles non-linear relationships between charging features and energy demand; works well with mixed engineered features and is suitable for tabular data.

Key parameters

Number of trees (n_estimators); maximum tree depth; minimum samples per split; random state.

Feature Engineering

Why chosen

Converts raw charging-session records into useful predictors for demand estimation.

Key parameters

Start hour; day of week; session duration; location/station; charger type (AC/DC).

Model Evaluation

Why chosen

Measures prediction error and goodness of fit to assess model performance.

Key parameters

MAE; RMSE; R^2 score; training/test split.

Selection criteria: Prediction accuracy, ability to model non-linear patterns, interpretability, implementation simplicity, and suitability for structured charging-session data.

Implementation / Model Building

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Feature Selection

- Used features: start hour, day of week, session duration, charger type (AC/DC), location (encoded), etc.
- Selected based on domain knowledge and feature importance from the Random Forest model.

Preprocessing Pipeline

- Handled missing values (median for numerical, mode for categorical).
- Encoded categorical features (One-Hot Encoding).
- Scaled numerical features using StandardScaler.
- Combined preprocessing steps using a Scikit-learn pipeline.

Data Splitting

- Train set : 70% of the data
- Validation set : 15% of the data
- Test set : 15% of the data
- Used random_state = 42 for reproducibility.

Configuration

- Model: Random Forest Regressor
- Key hyperparameters: n_estimators = 200, max_depth = 15, min_samples_split = 5, min_samples_leaf = 2
- Evaluation metrics: MAE, RMSE, R^2.

Code snippet (Python)

```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestRegressor

# Features and target
X = df.drop(columns=['energy_demand'])
y = df['energy_demand']

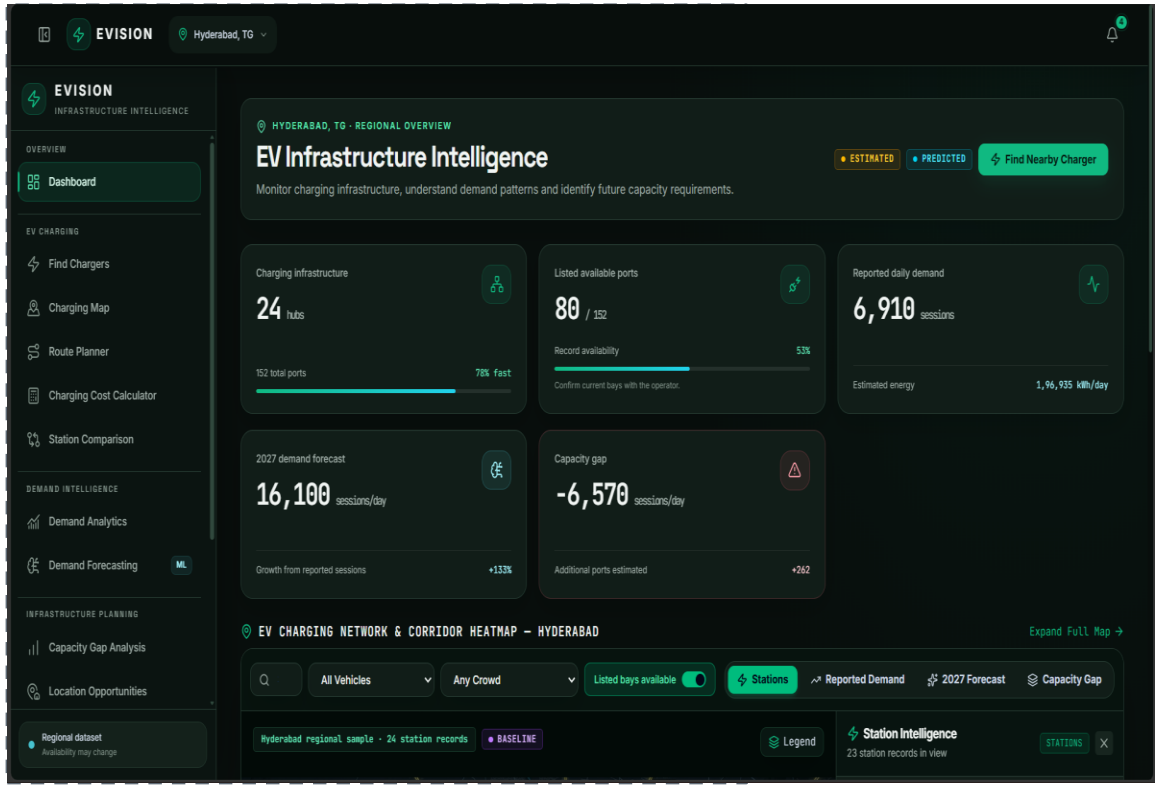
# Train-test split (70-15-15)
X_temp, X_test, y_temp, y_test = train_test_split(
    X, y, test_size=0.15, random_state=42
)
X_train, X_val, y_train, y_val = train_test_split(
    X_temp, y_temp, test_size=0.1765, random_state=42 # 0.1765 x 0.85 ≈ 0.15
)

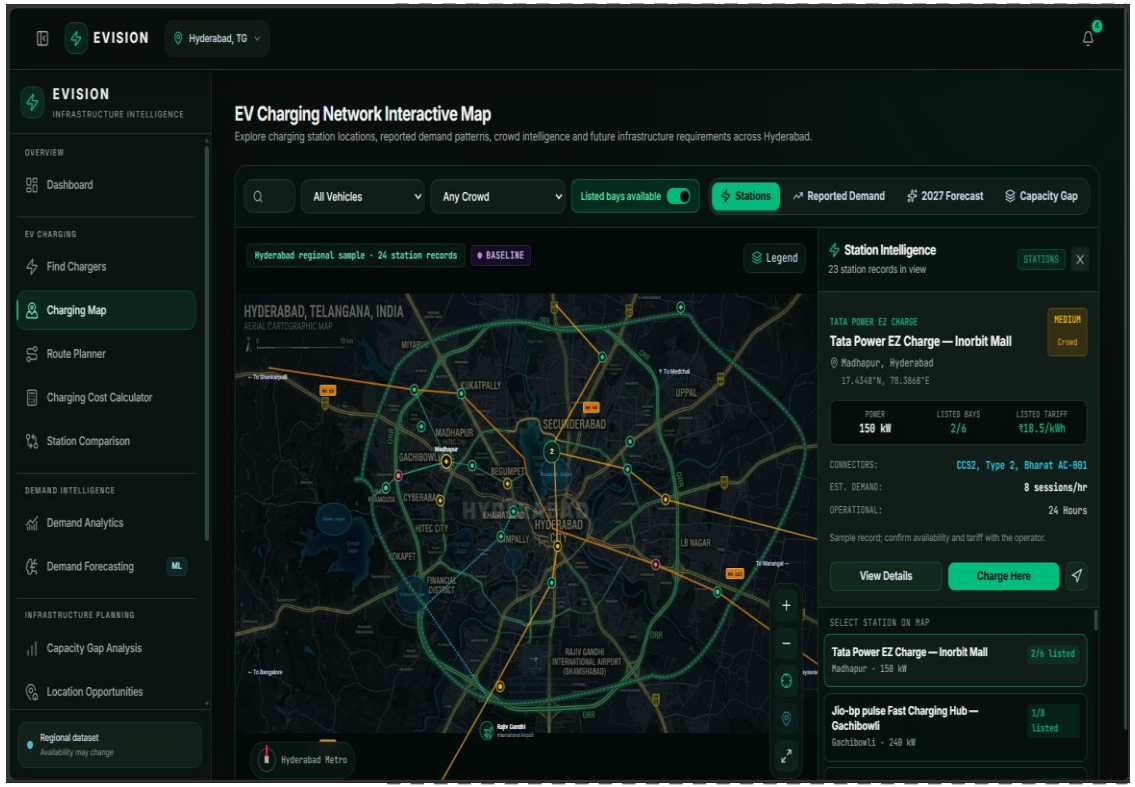
# Preprocessing pipeline
numerical_features = ['start_hour','session_duration']
categorical_features = ['day_of_week', 'charger_type', 'location']

preprocessor = ColumnTransformer([
    ('num', StandardScaler(), numerical_features),
    ('cat', OneHotEncoder(handle_unknown='ignore'), categorical_features)
])

# Model pipeline
model = Pipeline([
    ('preprocess", preprocessor),
    ('model', RandomForestRegressor(
        n_estimators=200,
        max_depth=15,
        min_samples_split=5,
        min_samples_leaf=2,
    ))  random_state=42
])
# Train the model
model.fit(X_train, y_train)
```

Working Prototype / Output Screens





Results & Performance Comparison

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Key Evaluation Metrics

MAE

Mean Absolute Error

2.14 sessions/hour

Average absolute difference between predicted and actual demand.

RMSE

Root Mean Squared Error

3.42 sessions/hour

Penalizes larger prediction errors more strongly.

R²

Coefficient of Determination

0.912

Indicates the proportion of demand variation explained by the model.

Model Evaluation Summary

Metric	Full Form	Reported Value
MAE	Mean Absolute Error	2.14 sessions/hr
RMSE	Root Mean Squared Error	3.42 sessions/hr
R²	Coefficient of Determination	0.912
MAPE	Mean Absolute Percentage Error	6.84%

Selected Model

Random Forest Regressor

Lower MAE and RMSE indicate smaller prediction errors; an R² score closer to 1 indicates a better fit.

Values shown are draft figures and should be checked against the final test-set output before submission.

Analysis & Key Findings

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Factors Used for Demand Prediction

The model uses time, session, charger and location information.

TIME

Start hour

Day of week

Captures daily and weekly demand patterns.

SESSION

Session duration

Charging activity

Represents the length of a charging session.

CHARGER

Charger type

AC / DC

Distinguishes charging equipment categories.

LOCATION

Hyderabad location

Station / area

Allows demand patterns to vary by place.

Input features are shown; individual feature-importance rankings are not available.

0.912

R² SCORE

Reported result for Random Forest Regressor

91.2% of target-demand variation is explained by the model.

Interpretation & Error Analysis

Prediction error	MAE: 2.14 sessions/hour
Larger-error sensitivity	RMSE: 3.42 sessions/hour
Model behaviour	Captures non-linear demand patterns
Key factors	Time • Duration • Charger type • Location

Note: These are draft figures from the project discussion; verify them against the final test output.

Testing & Validation

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Test ID	Test Description	Expected Result	Actual Result	Status
TC-01	Dashboard page loading	Dashboard and summary cards display correctly.	Dashboard loaded; summary cards displayed correctly.	**PASS**
TC-02	Hyderabad location filter	Selected area is applied and relevant station data appears.	Selected Hyderabad area was applied successfully.	**PASS**
TC-03	Demand prediction request	Predicted demand appears for the selected input values.	Demand prediction was generated and displayed successfully.	**PASS**
TC-04	Charging-station map	Map displays station markers and location labels.	Station markers and location labels appeared correctly.	**PASS**

✓ Functional Testing

4 / 4 checks passed

Dashboard loaded successfully. Location filter and map display worked as expected.

✓ Model Validation

Prediction check passed

Demand prediction was generated for the selected input values. Metrics: MAE, RMSE and R^2.

✓ Performance Testing

Basic operation passed

Application screens responded during manual testing. Formal load benchmarks were not run.

Result: All four functional test cases passed based on the completed manual checks.

Challenges & Limitations

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Challenges Faced & Solutions

01 Inconsistent source data

Standardized column names, timestamps and data formats before analysis.

02 Missing and duplicate records

Applied data-cleaning checks to handle incomplete and repeated entries.

03 Frontend–backend integration

Connected the React interface to FastAPI services through REST APIs.

Solution approach: clean data, standardize inputs and integrate application modules.

Project Limitations

01 Dataset coverage

Available records may not represent every EV charging station or all Hyderabad areas.

02 Historical-data dependence

Unexpected events and sudden demand changes may not be captured.

03 Live-data availability

Prediction quality depends on timely and accurate updates from the data source.

Future improvement: expand Hyderabad coverage and support more frequent data updates.

Conclusion

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01 EVision integrates historical EV charging data, data preprocessing and a Random Forest Regressor in a web application using React, FastAPI and MongoDB.

02 The project evaluation reports an R^2 score of 0.912, Mean Absolute Error (MAE) of 2.14 sessions/hour and Root Mean Squared Error (RMSE) of 3.42 sessions/hour.

03 The dashboard presents demand predictions and station-level information to support data-informed charging infrastructure planning across Hyderabad.

Final Outcome

0.912

R^2 SCORE

Random Forest Regressor

Objectives addressed: 4 of 4

Reported evaluation figure; verify against final test output.

Future Scope & Applications

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01 **Improve Prediction Performance**

Tune Random Forest parameters and compare additional regression models using MAE, RMSE and R^2.

02 **Expand Data & Features**

Add more charging-session records from different Hyderabad locations, with weather, traffic and calendar information.

03 **Deploy an Interactive Platform**

Host the dashboard online and add live station availability, demand alerts and downloadable analytical reports.

04 **Scale & Strengthen Security**

Introduce role-based access, secure data handling, regular backups and monitoring for reliable multi-user operation.

Applications

01 **Charging-station operators**

Plan charger allocation and prepare for high-demand periods.

02 **Urban & transport planners**

Identify areas that may need additional charging infrastructure.

03 **EV drivers and fleet managers**

Use demand insights to plan charging activities and fleet schedules.

04 **Charging-network providers**

Monitor station-level usage and support capacity-expansion decisions.

Potential adopters: charging networks, fleets and city infrastructure teams.

Team Contribution

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Registration No.	Team Member	Contribution / Modules Handled
2420030147	A. Ramcharan Reddy	• Data collection and preprocessing; exploratory data analysis. • Feature preparation, Random Forest model development and evaluation. • Research survey and technical documentation.
2420030760	P. Chaitanya Kumar	• Backend development using FastAPI and MongoDB integration. • REST API design, data flow and prediction-service integration. • System architecture, testing and technical documentation.
2420030281	Y. Durga Prasad Reddy	• Frontend development using React and TypeScript. • Dashboard design, demand visualizations and Hyderabad station map. • Frontend–backend integration, validation and presentation.

Teamwork & Coordination

All three members contributed to requirement discussions, integration reviews, final validation and project presentation.

References

[1] K. Baek, E. Lee, and J. Kim, “A dataset for multi-faceted analysis of electric vehicle charging transactions,” *Scientific Data*, vol. 11, Article 262, 2024. doi: 10.1038/s41597-024-02942-9.

[2] X. Wang et al., “Charging load forecasting of electric vehicles based on sparrow search algorithm-improved random forest regression model,” *The Journal of Engineering*, 2023, doi: 10.1049/tje2.12280.

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[4] Microsoft, “TypeScript Documentation.” Available: https://www.typescriptlang.org/docs/

[5] Vite, “Vite Guide and Documentation.” Available: https://vite.dev/guide/

[6] Tailwind Labs, “Tailwind CSS Documentation.” Available: https://tailwindcss.com/docs

Project Links

GitHub URL:
https://github.com/chaitanyakumar00/DS_S4_Team20_EVision/tree/main/Project

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THANK YOU

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